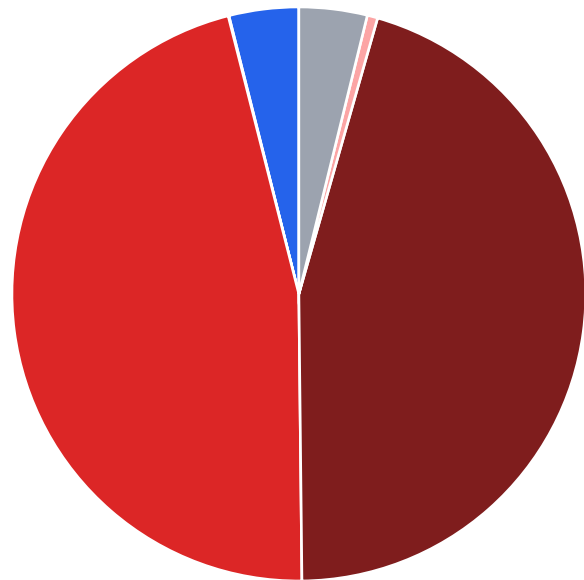


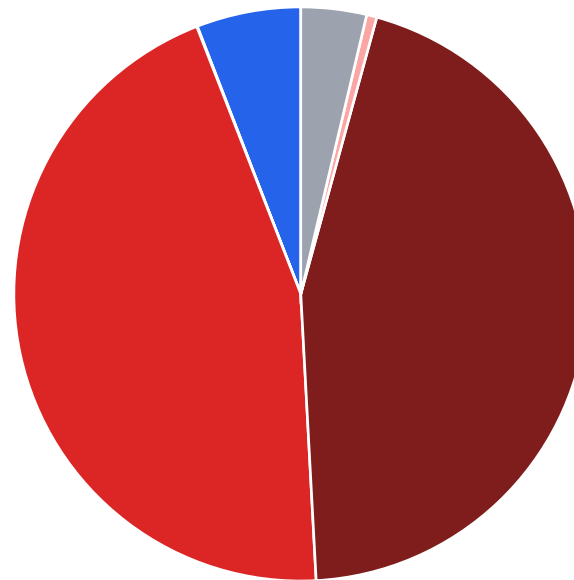
## Time distribution in realistic modes

Baseline + Overhead  
(total:  $167.3 \pm 6.3$  ms, n=22)



| Component   |       |           |
|-------------|-------|-----------|
| Algorithm   | 3.9%  | (6.6 ms)  |
| Auth (JWT)  | 0.0%  | (0.1 ms)  |
| DB read     | 46.2% | (77.3 ms) |
| DB write    | 45.4% | (76.0 ms) |
| DB rollback | 0.6%  | (1.0 ms)  |
| Other       | 3.8%  | (6.4 ms)  |

Full + Overhead  
(total:  $173.1 \pm 10.2$  ms, n=22)



| Component   |       |           |
|-------------|-------|-----------|
| Algorithm   | 5.9%  | (10.2 ms) |
| Auth (JWT)  | 0.0%  | (0.1 ms)  |
| DB read     | 44.9% | (77.8 ms) |
| DB write    | 44.9% | (77.7 ms) |
| DB rollback | 0.6%  | (1.0 ms)  |
| Other       | 3.7%  | (6.4 ms)  |